

Formulas (all valid LaTeX):

$$\frac{\partial L}{\partial \theta} = \sum_{i=1}^n x_i - \mu$$

$$\int_a^b f(x) \, dx = F(b) - F(a)$$

$$\sqrt{x^2 + y^2} = \sqrt{r^2}$$

$$\binom{n}{k} = \frac{n!}{k!(n-k)!}$$

Inline formulas in prose:

The function $f(x) = \alpha x + \beta$ is linear.

The limit $\lim_{n \rightarrow \infty} 1/n = 0$.